JULIEN RABAULT

AI Engineer, LLM agents and RAG in production

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I design agent harnesses and ship LLM agents and RAG systems that run in production at customer sites, after four years of deep learning on a national supercomputer at CNRS (PyTorch, multi-GPU). Two peer-reviewed publications.

SKILLS

Agentic systems: Agent harnesses (agent loop, tool calling, middleware, memory, human-in-the-loop), LangChain / LangGraph / Deep Agents, RAG, MCP, evals (golden sets, LLM-as-judge, A/B testing), multimodal (OCR, vision, audio), fine-tuning

Languages & data: Python, PyTorch, FastAPI, SQL, Weaviate, PostgreSQL, MongoDB, Hugging Face, Mistral and OpenAI APIs

Infrastructure: Docker, GitLab CI, AWS, Celery, RabbitMQ, Airflow, Langfuse, Slurm, Linux

PROFESSIONAL EXPERIENCE

AI Engineer, Berger-Levrault, Toulouse (AI R&D team)

Jan. 2026 - present

Athena, agentic platform wired into its clients' documents and business APIs (local government, legal, industry, public-sector HR): 100k documents, 6-month pilot with 70 users, production rollout under way. Needs gathered from business units, client workshops.

Platform core

- **Agent harness** (LangChain / LangGraph), architecture designed and owned. Moved from a router (one agent per task) to a single agent loop configured per profile and extended through middleware: persistent memory, human approval of sensitive actions, context management (summarisation, token budget), guardrails, skills, sub-agents. Prompt caching shared across conversations, tenants and clients: inference costs cut by an estimated 40-50%. Now migrating to Deep Agents.
- **Agent tools**: agentic RAG, web search, calculator and the matching MCP servers, all plugged into the same loop. Adding a tool no longer means building a new graph.
- **Evaluation and monitoring**: rebuilt monitoring on Langfuse v4 (traces, datasets, automated evaluations and scoring, LLM-as-judge), golden sets, model, prompt and architecture comparisons, regression and cost tracking; sandboxing of retrieved content against indirect prompt injection.
- **MCP Builder**: turns internal business APIs into MCP servers. An LLM selects the useful endpoints, audits their gaps and generates purpose-built tools (workflows or Python code via FastMCP), human-in-the-loop.

LLM services and pipelines

- **Content-extractor**, multimodal extraction (OCR / PDF / DOCX / audio): FastAPI service built from scratch, 7 pipelines, async batch processing (Celery + Mistral batch API), factory/registry architecture. Deployed, -50% on extraction costs.
- **Template agent**: extracts templates from Word documents that a second agent fills from natural language. In production.
- **Business data structuring** (public-sector HR): interview write-ups turned into normalised, deduplicated needs matched against a catalogue. Enriched RAG ingestion (Airflow): chunking, embeddings, indexed questions.

Machine Learning Engineer, CNRS (PNRIA), Toulouse

Dec. 2021 - Jan. 2026 (4 yrs)

Engineer embedded in research teams (weather, astrophysics, materials, ethology, biology): 10+ projects of 6-12 months, two in parallel, from requirements to delivery, for Météo France, CNES, CEA and INEE. Training and fine-tuning on the Jean Zay supercomputer (multi-GPU DDP, Slurm). Created and taught "Introduction to LLMs" to 25 PhD students.

Selected projects

- **GENS / MetScore, Météo France:** multi-GPU training and fine-tuning of a diffusion model (DDPM) in PyTorch; designed MetScore (metrics library), still in production. POC: -20% compute, equivalent quality. Co-author, AIES 2025.
- **DeepFaune, CNRS / INEE:** fine-tuned YOLOv8 on 1.5M images (24 classes) with class-imbalance handling. 93% accuracy, 3x faster inference. Peer-reviewed publication.
- **BIGSF, CNES:** tech lead on the rebuild of a U-Net galactic image analysis library, public toolbox.
- **AUTOFILL, CEA:** PairVAE for nanomaterial data completion (0.98 MAE). **MORPHOGAN, Univ. Lorraine:** StyleGAN2 codebase rebuild, automated and containerised pipeline.

Software Engineer, Agileo Automation, Montauban

Aug. 2020 - Sept. 2021 (1 yr)

- **Supervision framework for manufacturing equipment** (semiconductors): C#, HMI, CI/CD, team of 5 in Scrum.

EDUCATION

MSc Artificial Intelligence & Pattern Recognition, Université Paul Sabatier / IRIT (2021); **BSc Computer Science**, Université Paul Sabatier (2019)

PUBLICATIONS & OPEN SOURCE

Enriching Operational High-Resolution Ensemble Forecasts with StyleGAN-2, AIES 2025, peer-reviewed.

The DeepFaune initiative: automatic identification of European fauna, peer-reviewed.

LangChain: merged contribution (Mistral integration); **LLMock** (PyPI): LLM mock server, 10+ providers; **DDPM-weather**: diffusion for weather images.